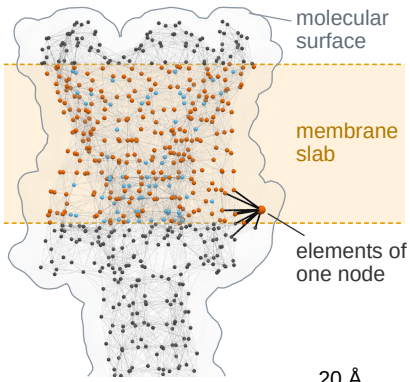
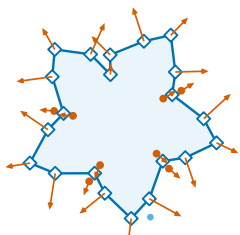
Cα outside slab · membrane Cα · loaded outer-shell node · polygon vertex · nodal force $\tau\,\partial A/\partial \mathbf{x}$

a MscL Cα truss, r_c = 10 Å



b One 4 Å layer



c Formulation and workflow

Geometrically nonlinear truss
$U = (k/2)\,\Sigma_e(\ell_e - \ell_e^0)^2$, exact kinematics

↓

Conservative membrane tension
$\Pi = U - \tau A(\mathbf{x})$, $A = \mathbf{x}^T\mathbf{H}\mathbf{x}/2$

↓

Gauge from the load symmetries
A invariant to 4 rigid motions; tilt has a moment

↓

Verified path, stability by inertia
Newton = VFIFE; inertia of $\mathbf{K}_t - \tau\mathbf{H}$ each step

↓

Departure from linearization
stiffening a, turning $\sin\theta \approx \chi$, snapping n_{neg}

↓

Test against partner structures
TM Cα RMSD vs. linear response, null models